

Tensor Products on Quantum Spin Systems

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In the 1960s Friedrichs met Heisenberg, and used the occasion to express to him the deep gratitude of the community of mathematicians for having created quantum mechanics, which gave birth to the beautiful theory of operators in Hilbert space. Heisenberg allowed that this was so; Friedrichs then added that the mathematicians have, in some measure, returned the favor. Heisenberg looked non-committal, so Friedrichs pointed out that it was a mathematician, von Neumann, who clarified the difference between a self-adjoint operator and one that is merely symmetric. “What’s the difference,” said Heisenberg.

I. INTRODUCTION - WHAT IS A PHYSICAL SYSTEM?

The question of what a physical system is goes back to the initial **DO BETTER**. Essentially, we require a C^* algebra, which shall describe the possible operators that can act on a system. We call the self adjoint elements of this C^* algebra the observables. To connect this to the standard Hilbert space representation taught in a first course in quantum mechanics, we can look at *representations* of the C^* algebra. We already covered most of these

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II. THE QUASI-LOCAL ALGEBRA

III. DYNAMICS OF A PHYSICAL SYSTEM

A. What Are Dynamics?

In quantum mechanics, we implement the dynamics of a quantum system in the Heisenberg representation, where all observables have the time dependence

$$A(t) = U(t)^* A U(t)$$

where $U(t) = e^{-iHt}$ is the unitary time evolution operator generated by the Hamiltonian H . However, this leaves a major question. We know that H is, in general, unbounded. Thus, we should not expect $U(t)$ to be well defined, or a member of the Hilbert space at all. Can we rescue it?

First, let us suppose that a system already has dynamics defined on it. We shall define this as follows:

Definition 1. A C^* dynamical system is a C^* algebra $\mathcal{A}$ together with a strongly continuous 1 parameter semigroup $t \mapsto U(t)$, inducing an isomorphism $\alpha_t : \mathcal{A} \rightarrow \mathcal{A}$ with $\alpha_t : A \mapsto U(t)^* A U(t)$.

Now, if we were able to write this automorphism in terms of a unitary operator $U(t)$ such that $\alpha_t(A) = U(t)^* A U(t)$, we would be able to apply Stone's theorem:

Theorem 2 (Stone). Let $t \mapsto U(t)$ be a 1 parameter strongly continuous semigroup of unitary operators acting on some Hilbert space $\mathcal{H}$. Then there exists some a unique (possibly unbounded) operator H such that

$$U(t) = e^{itH}.$$

We do this, in essence, by writing the automorphism in the Schrödinger picture.

Theorem 3. Let $\mathcal{A}$ be a C^* subalgebra of $\mathcal{B}(\mathcal{H})$, and $\alpha : \mathcal{A} \rightarrow \mathcal{A}$ an automorphism of $\mathcal{A}$. Then there is a unitary U such that $\alpha(A) = UAU^*$ for all $A \in \mathcal{A}$.

Proof. We define the unitary U by $U : \mathcal{AH} \rightarrow \mathcal{H}$ by $UA|\psi\rangle = \alpha(A)|\psi'\rangle$, where $|\psi\rangle$ is defined by $\alpha(|\psi\rangle\langle\psi|) = |\psi'\rangle\langle\psi'|$.

First, we note that U is linear, which follows by using the automorphism properties of $\mathcal{B}(\mathcal{H})$ to pull out sums on the right hand side of the equation. Thus, U is well defined.

We then prove that U is unitary. Let $|\phi\rangle, |\psi\rangle \in \mathcal{H}$. We consider the subspace of operators $|\phi\rangle\langle\phi|$ and $|\psi\rangle\langle\psi|$. Since these are rank 1 operators, the proof transfers over from finite dimensional linear algebra: We have

$$\text{tr}(|\psi\rangle\langle\psi| |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|^2$$

Since automorphisms preserve the trace, we have

$$|\langle\psi|\phi\rangle|^2 = |\langle\psi'| \phi'\rangle|^2 = |\langle\psi|U^*U|\phi\rangle|^2$$

and thus U is unitary. Finally, since **TODO**

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